A Survey of Artificial Intelligence in Multiple Sclerosis MRI: Datasets, Deep Learning Methods, Challenges, and Future Directions

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***Abstract*— Magnetic resonance imaging (MRI) is central to multiple sclerosis (MS) diagnosis and monitoring, but manual interpretation is time-consuming and poorly suited to reproducible lesion quantification, multimodal integration, and longitudinal comparison. This survey evaluates artificial intelligence for MS MRI through a connected chain of data realism, architectural mechanism, clinical task, measurement, evidence quality, and translation. Public challenges have enabled reproducible white-matter and new-lesion segmentation, yet remain small, brain-centered, and incompletely representative of routine practice. Convolutional encoder–decoders provide an established technical foundation, while nnU-Net supplies a reproducible configuration baseline rather than clinical validity. Vision Transformer, Swin Transformer, UNETR, and CNN–Transformer hybrids expand global modeling but require more data and computation, whereas generative, state-space, self-supervised, and foundation models remain less clinically established. Training objectives must reflect the target error: overlap losses address imbalance, boundary-aware objectives refine geometry, reconstruction objectives preserve image fidelity, and survival losses accommodate censoring. Across tasks, high internal Dice or area under the curve does not establish transportability because scanner shift, label uncertainty, patient leakage, spectrum bias, absent calibration, and weak external validation remain common. Lesion segmentation, quantitative analysis, and longitudinal change detection are closest to assistive clinical use; autonomous diagnosis and individual prognosis are not. Progress now depends on multicenter prospective evaluation, pathology-preserving generation, calibrated uncertainty, equitable data governance, workflow-aware endpoints, and monitored human–AI deployment.**

***Index Terms*— Artificial intelligence, deep learning, magnetic resonance imaging, multiple sclerosis**

I. Introduction

Multiple sclerosis is a chronic inflammatory and neurodegenerative disorder with marked variation in lesion location, relapse activity, tissue loss, cognition, and disability trajectory. MRI supports dissemination in space and time, exclusion of mimics, treatment monitoring, and characterization of occult tissue injury. The 2024 revisions of the McDonald criteria and the accompanying MAGNIMS–CMSC–NAIMS recommendations further elevate imaging by recognizing the optic nerve and incorporating the central vein sign (CVS) and paramagnetic rim lesions (PRLs) under specified conditions [1], [2]. Earlier consensus statements established the acquisition and interpretation principles on which these changes depend [3], [4]. The practical consequence is an expanding reading task: clinicians must combine brain, spinal cord, and sometimes optic-nerve examinations; compare several contrasts and prior visits; distinguish new inflammatory activity from technical variation; and place imaging findings in a clinical differential diagnosis.

Manual review is therefore limited less by a lack of expertise than by scale and reproducibility. Small lesions are easily missed, confluent lesions are difficult to count, volume measurements vary with contouring policy, and serial comparison is sensitive to positioning, resolution, and registration. Susceptibility markers require additional lesion-level assessment. Quantitative measures of atrophy are below the threshold of reliable visual estimation over short intervals. Artificial intelligence (AI) can assist at each stage: reconstruction, quality control, registration, missing-sequence synthesis, harmonization, lesion segmentation, biomarker detection, diagnosis, and prognosis. These functions are not interchangeable. A method that generates visually convincing FLAIR images may still erase a 3-mm lesion; a high Dice score does not establish safe longitudinal monitoring; and a classifier that separates established MS from healthy controls does not reproduce diagnosis among patients with plausible mimics.

Deep learning in MS MRI has progressed from patch-based convolutional neural networks (CNNs) to multiscale encoder–decoders, self-configuring pipelines, Transformers, generative models, and large pretrained representations. Existing reviews document this growth, but pooled accuracy estimates and model-by-model catalogs can hide the determinants of clinical value: unique patients rather than image counts, patient-level splitting, annotation uncertainty, acquisition shift, calibration, external validation, and the decision supported [5], [6]. Fig. 1 therefore frames the survey as a chain from imaging input and data evidence to architecture, task output, measurement, and clinical evidence. It also defines the survey's three contributions: a data-realism analysis of MS benchmarks, a mechanism-centered synthesis of deep networks and training objectives, and a task-specific appraisal that separates technical performance from

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clinical utility. The organizing question is not which network is universally best, but which inductive bias, training objective, data regime, and validation design fit a particular MS MRI task.

This is a structured critical survey rather than a registered systematic review. We examined publications from January 2015 through June 2026, supplemented by foundational architectures, challenges, diagnostic criteria, and regulatory sources. PubMed/MEDLINE, IEEE Xplore records, publisher pages, challenge repositories, public device records, and backward and forward citation tracking were used. Sources were included when they described an MS-relevant dataset, AI method, technical or external validation, workflow evaluation, or applicable reporting and deployment guidance. Unverifiable scraped collections, non-imaging AI, and studies without separable training and testing evidence were excluded. We appraised the evidence using CLAIM 2024, TRIPOD+AI, PROBAST+AI, STARD-AI, DECIDE-AI, and FUTURE-AI [7]–[12]. No meta-analysis was performed because lesion definitions, prevalence, sequences, test populations, and validation levels were too heterogeneous for defensible pooling. This approach supports a task-level and evidence-level synthesis while acknowledging that it is not exhaustive.

II. Fundamentals of MRI Images and Public Benchmarks

A. MRI Images as Inputs for Artificial Intelligence

For AI, an MRI examination is best understood as a spatially registered collection of voxel arrays whose contrasts provide complementary, imperfect views of tissue. T1-weighted images describe anatomy and support tissue segmentation and atrophy measurement; T2-weighted and proton-density images increase sensitivity to water-rich abnormalities; FLAIR suppresses cerebrospinal-fluid signal and is the workhorse for supratentorial white-matter lesions; post-contrast T1-weighted imaging reveals blood–brain-barrier disruption; double inversion recovery can increase cortical-lesion conspicuity; and susceptibility-weighted imaging or quantitative susceptibility mapping supports CVS and PRL assessment [13]–[16]. These contrasts are not labels for fixed biological states. A lesion may be visible on one sequence and ambiguous on another, enhancement is time-limited, and susceptibility markers depend on acquisition and reader eligibility rules. Fig. 2 links these inputs to the MS targets they can support without implying that every sequence is available in routine care.

Brain MRI is commonly represented as a stack of axial slices or a three-dimensional volume; spinal cord MRI adds a small, elongated structure affected by partial volume, motion, pulsation, and inconsistent axial coverage. Two-dimensional networks are inexpensive and can exploit large in-plane resolution, but ignore through-plane context. A 2.5-D method combines neighboring slices or orthogonal views while retaining two-dimensional operators. Three-dimensional networks model volumetric continuity and permit voxel-level outputs, but their receptive field is constrained by patch size, memory, and anisotropic resolution. Multisequence networks can stack aligned contrasts as channels, use contrast-specific encoders, or fuse features later. Each choice imposes assumptions about sequence availability and registration. A model trained on T1, T2, and FLAIR cannot silently treat a missing channel as an ordinary zero image.

MRI intensity has no universal physical scale across scanners. Field strength, manufacturer, receive coil, pulse sequence, reconstruction software, patient positioning, and software upgrades can all alter contrast and texture. Preprocessing—bias-field correction, skull removal, denoising, registration, resampling, intensity normalization, and cropping—can reduce irrelevant variation, but may also introduce shortcuts or suppress pathology. Resampling changes the apparent size of small lesions; aggressive registration can deform new lesions; normalization based on all study data leaks test-distribution information; and harmonization can remove disease signal when site and phenotype are correlated. DeepHarmony and HACA3 illustrate learned contrast and site harmonization, whereas super-resolution can improve apparent detail in thick-slice FLAIR, but each transformation requires downstream lesion-preservation tests rather than PSNR alone [17]–[19]. Augmentation should similarly represent plausible variation in geometry, intensity, noise, motion, and missing contrasts without deleting or inventing clinically meaningful lesions.

Table I summarizes the principal relationship between sequences and targets. The central design principle is conditional sufficiency: inputs should contain the information needed for the intended output, and evaluation should match the conditions under which those inputs will be acquired. A FLAIR-only model may be suitable for screening new white-matter lesions but not for classifying enhancement, CVS, or PRL. Conversely, a specialized susceptibility model may be technically strong but operationally limited by sequence availability and non-evaluable lesions.

B. Public MS MRI Benchmarks

Public challenges made MS imaging AI measurable, but scan counts can exaggerate independent sample size. The ISBI 2015 longitudinal resource contains 82 sessions from only 19 patients, all acquired in one 3-T Philips environment; its repeated visits are useful for longitudinal analysis but create severe leakage if sessions rather than patients are randomized [20], [21]. MSSEG 2016 contains 53 patients across three sites and four scanners, with seven expert delineations and an unseen-scanner component in testing [22], [23]. Its enduring value is the explicit demonstration that the reference standard changes with the rater and fusion rule. MSSEG-2 advances from cross-sectional burden to new-lesion segmentation in 100 longitudinal patient pairs from 15 scanners. Approximately 40% of lesions in the final reference were initially identified by no more than two of four readers and then underwent senior-expert adjudication; this is evidence of concentrated reader difficulty, not evidence that expert annotation is generally unreliable [24].

Other public sets broaden geography and acquisition while remaining individually modest. The 3D-MR-MS collection contains 30 single-site 3-T cases with four contrasts and consensus masks; Long-MR-MS contains 20 patients and 40 older 1.5-T examinations with labels for new, disappearing,

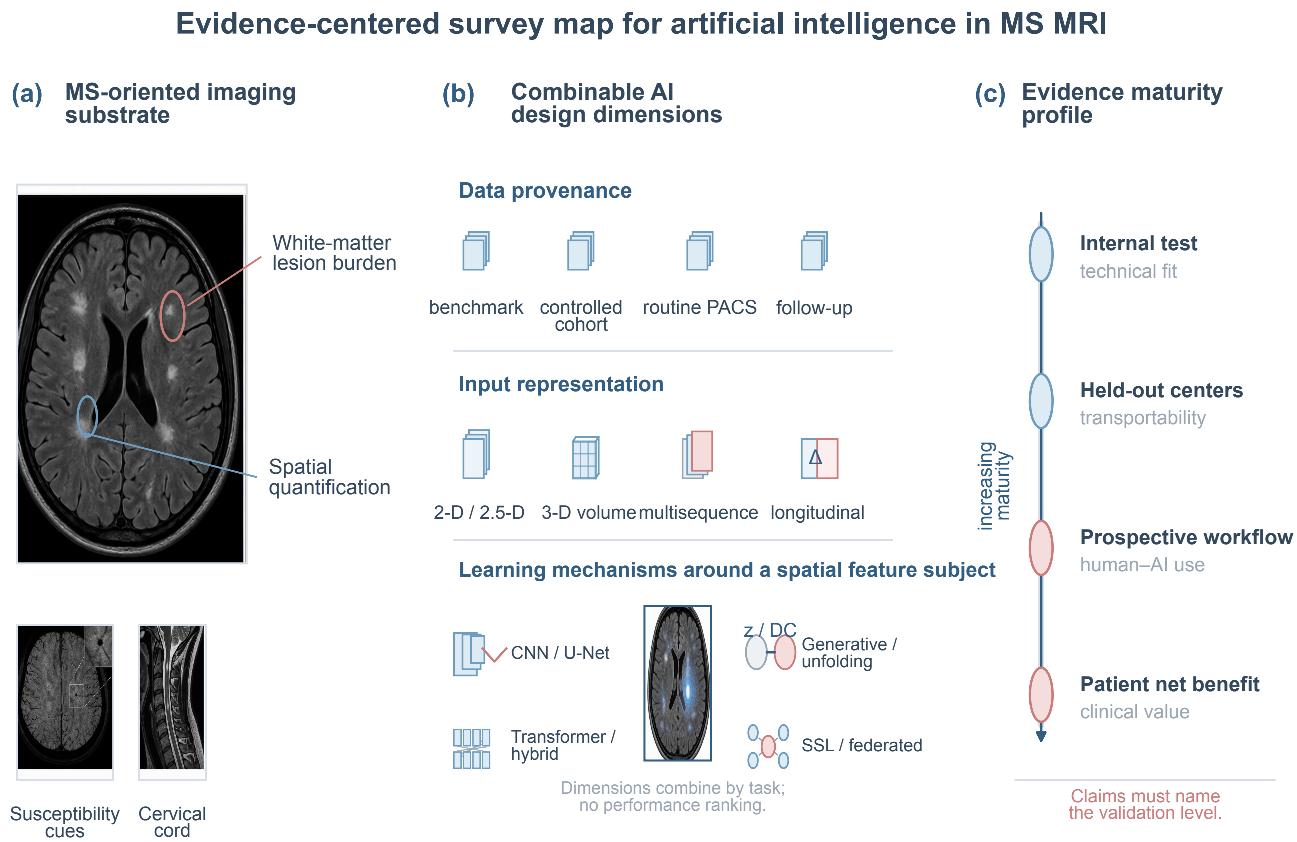

Fig. 1. Evidence-centered survey map. Synthetic FLAIR, susceptibility-sensitive, and cervical-cord MRI provide the imaging substrate; data provenance, input representation, and learning mechanism are orthogonal design dimensions that combine according to task; and the evidence profile separates internal testing, held-out centers, prospective workflow evaluation, and patient net benefit. The single vertical arrow denotes increasing validation maturity, not increasing model performance.

TABLE I
MRI SEQUENCES, AI INPUTS, AND MS TARGETS

MRI input	Typical AI representation	MS-relevant target	Primary limitation
T1-weighted	2-D/3-D anatomy channel	Tissue segmentation, regional volume, atrophy	Lesions may be subtle; scanner-dependent contrast
T2-weighted / PD	Single or paired channels	Water-rich lesions and cord abnormalities	CSF and nonspecific hyperintensity reduce specificity
FLAIR	2-D, 2.5-D, or 3-D volume	White-matter burden; new or enlarging lesions	Motion, thick slices, confluent lesions, and vascular mimics
Post-contrast T1	Registered pre/post or post-only volume	Active enhancing lesions	Rare positives, vessels, and acquisition timing
DIR / PSIR	High-resolution cortical contrast	Cortical and leukocortical lesions	Limited availability and high rater uncertainty
SWI / T2* / QSM	Magnitude/phase or susceptibility maps	CVS and PRL	Eligibility rules, orientation, and specialized acquisition
Spinal T2/STIR/PSIR	Sagittal plus axial 2-D/3-D inputs	Cord area and intramedullary lesions	Small anatomy, pulsation, partial volume, and incomplete coverage

enlarging, and shrinking lesions [25], [26]. MSLesSeg provides 115 examinations from 75 patients across several systems; MS-Baghdad contributes 60 heterogeneous routine examinations from 20 centers [27], [28]. SibBMS combines MS or radiologically isolated syndrome cases and controls, but its public documentation should not be interpreted as showing dense manual masks for every participant [29]. MS3SEG distinguishes MS lesions from other white-matter hyperintensities in 100 single-scanner cases but excludes lesions below 3 mm [30]. PediMS has only nine children and 28 scans, making it more defensible as a difficult out-of-distribution probe than a training corpus [31]. Shifts 2.0 repackages several existing resources for uncertainty research and therefore must not be counted as an independent cohort [32].

Spinal data remain much less accessible than brain FLAIR. A 30-site study of 1,042 participants supported robust cord segmentation but markedly lower lesion agreement, illustrating that cord anatomy and intramedullary lesions are distinct tasks [33].

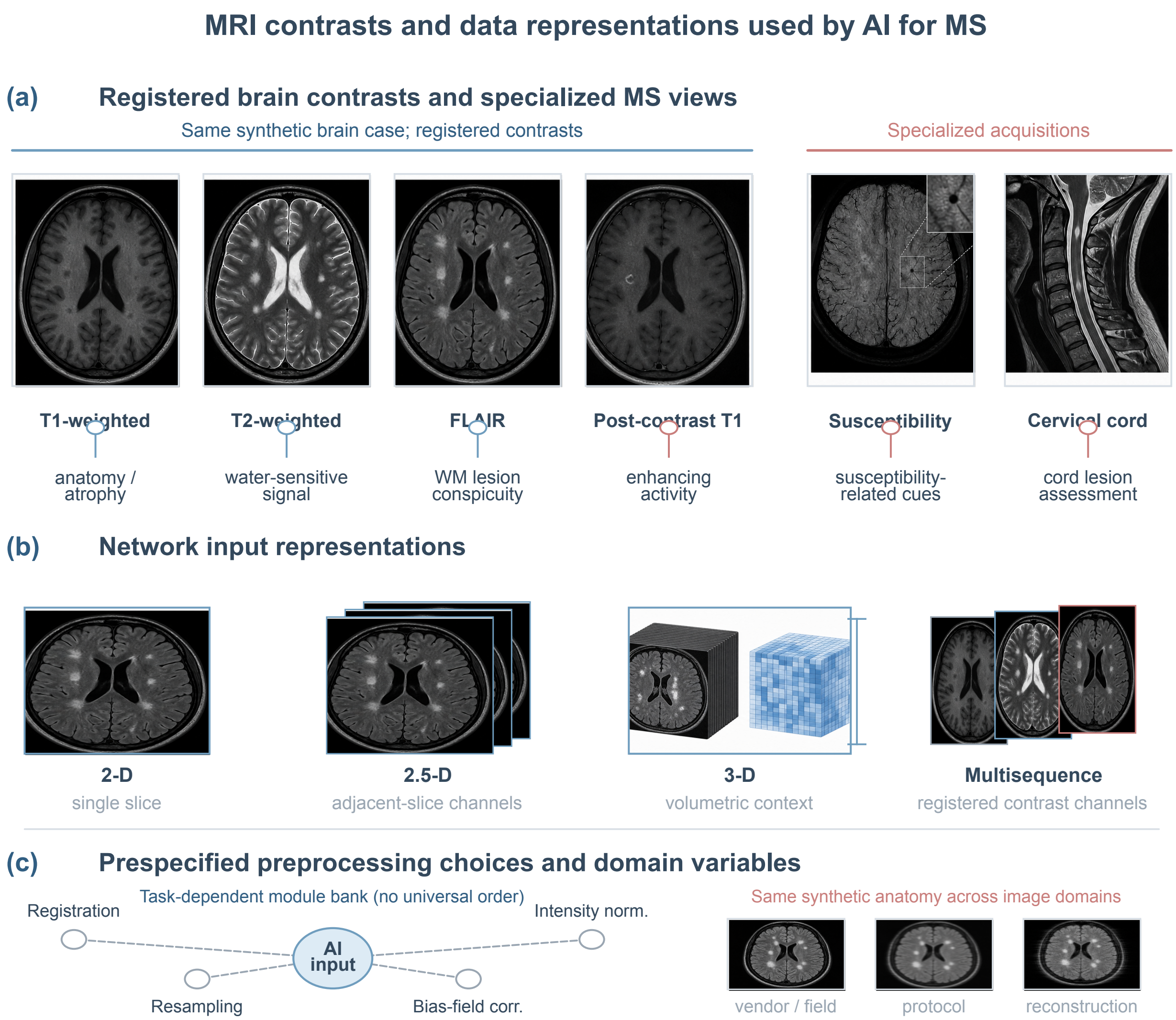

Fig. 2. MRI inputs for MS-oriented AI. Synthetic schematic anatomy illustrates complementary T1-weighted, T2-weighted, FLAIR, post-contrast T1-weighted, susceptibility-sensitive, and cervical-cord appearances; it is not patient data. Vector annotations connect sequence information to white-matter lesions, enhancement, CVS, PRL, cord lesions, 2-D/2.5-D/3-D representations, preprocessing, and acquisition-domain variables.

Recent multicenter work increases contrast and center diversity, yet its 4,428 annotated MRI images are the source-reported unit and must not be relabeled as independent examinations or 3-D volumes; the trained model is public, whereas the constituent MRI are not a reusable public benchmark [34]. This asymmetry has shaped research: white-matter segmentation appears mature partly because it has labels and leaderboards, whereas cortical lesions, enhancement, CVS, PRL, spinal lesions, cognition, and treatment response have fewer shared test sets.

Controlled cohorts and clinical-trial archives provide complementary evidence. The Swiss Multiple Sclerosis Cohort and OFSEP link prospective clinical and imaging follow-up across specialist centers, although care-driven missingness and protocol substitutions remain [35]–[37]. MS PATHS enrolled 16,568 participants; 8,364 participants contributed 14,414 standardized 3-T MRI studies. Those counts should not be conflated, and the Siemens-specific acquisition did not provide dense lesion labels at cohort scale [38]. The NAIMS 7-T repository supports chronic-active and cortical-lesion research in a specialized high-field population [39]. MindGlide used 4,247 development scans from 592 scanners and externally evaluated 14,952 scans from 1,001 people and 186 scanners. Its development-patient count is recorded as NR (source discrepancy) because the abstract reports 2,934 whereas the Methods report 2,871; the scale and scanner diversity remain informative, but proprietary trial composition limits independent reproduction [40]. Fig. 3 distinguishes open download and registration-controlled benchmarks from application-controlled cohorts and proprietary archives. Table II reports access mode separately from license status and places these sources on a control–realism continuum rather than ranking them by size.

C. Dataset Bias and Benchmark Realism

A benchmark is clinically realistic only when its population, acquisition, labels, and evaluation mirror the intended use. Common failures include randomizing patches or visits instead

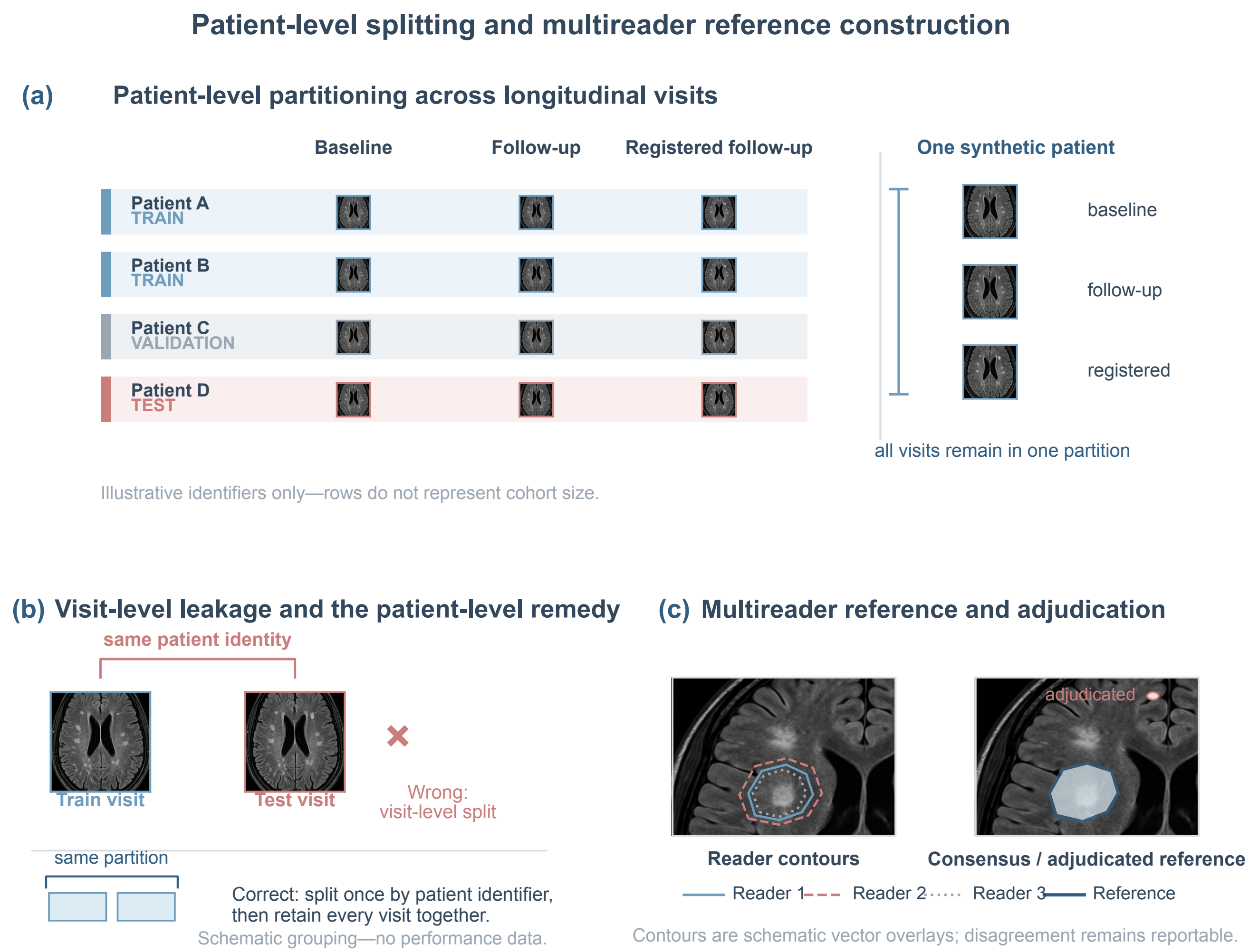

Fig. 3. Patient-level partitioning and multireader reference construction. Every visit from one patient is assigned to a single training, validation, or test partition; placing baseline and follow-up visits across partitions creates identity leakage. Reader-specific vector contours on synthetic FLAIR anatomy are retained before consensus or adjudication so that disagreement remains auditable. Patient letters and rows are illustrative identifiers, not a cohort-size statement.

of patients; using healthy controls acquired through a different pipeline; tuning repeatedly against a public test server; learning scanner or skull-stripping artifacts; and reporting only aggregate Dice in cohorts dominated by large lesions. Annotation is itself a measurement process. Raters disagree about punctate lesions, confluent boundaries, cortical location, enhancement, and susceptibility markers. Consensus can improve reproducibility but can also hide uncertainty; probabilistic or multirater labels preserve information that a binary fused mask discards.

Dataset size does not repair spectrum bias. A large trial archive can underrepresent comorbidity, advanced disability, nonadherence, or low-resource scanners. A heterogeneous PACS archive contains clinically useful variation but also informative missingness, report-label errors, and protocol decisions that encode clinicians' prior beliefs. Patient-level splitting, site-held-out testing, demographic reporting, negative cases, scanner-upgrade stress tests, and locked preprocessing are therefore minimum safeguards. Performance on a hidden benchmark is valuable technical evidence, but it remains one step in an evidence ladder rather than proof of general clinical behavior.

Benchmark reuse adds a subtler form of overfitting. Even when masks remain hidden, repeated submissions reveal enough information to guide architecture and threshold choices toward a fixed test distribution. A leaderboard should therefore control submissions, preserve a sequestered final set, and refresh acquisition conditions over time. Data cards should state the number of unique patients, visits per patient, sites, scanners, field strengths, native and resampled resolution, sequence availability, disease phenotype, treatment, age and sex distribution, rater process, license, and exclusions. "Multicenter" is not a sufficient descriptor when one dominant scanner or institution supplies most cases. Likewise, a public download is not necessarily open science if its license, consent scope, or derivative-use conditions are ambiguous.

TABLE II
MS MRI DATASETS, BENCHMARKS, AND CONTROLLED COHORTS

Resource	Independent people / scans	Sequences and task	Centers / scanners	Longitudinal	Reference standard	Access mode	License	Main limitation
ISBI 2015 [20], [21]	19 / 82	T1, PD/T2, FLAIR; white-matter lesions	1 site; 3-T Philips	4–6 visits	Two manual raters	Registration-controlled	NR	Only 19 people; visit-level leakage risk
MSSEG 2016 [22], [23]	53 / 53	T1, FLAIR, PD/T2; white-matter lesions	3 sites; 4 scanners	No	Seven experts and fusion	Registration-controlled	NR	Small labeled training set; target depends on fusion rule
MSSEG-2 [24]	100 / 200	Paired FLAIR; new lesions	15 scanners; 3 vendors	Two visits	Four readers plus senior adjudication	Registration-controlled	NR	FLAIR-only; difficult lesions concentrate disagreement
3D-MR-MS [25]	30 / 30	T1, T2, FLAIR, post-Gd T1; lesions	1 site; 3-T Siemens	No	Three-expert consensus	Open download	NR	Small single-scanner cohort
Long-MR-MS [26]	20 / 40	T1, T2, PD, FLAIR; lesion change	1 site; 1.5-T Philips	Two visits	Two-expert consensus	Open download	NR	One interval and 3-mm sections
MSLesSeg [27]	75 / 115	T1, T2, FLAIR; white-matter lesions	Several 1.5-T systems	1–4 visits	One rater; three-expert review	Open download	NR	Most participants have one visit
MS-Baghdad [28]	60 / 60	Routine T1/T2/FLAIR; lesions	20 centers; 1.5 T	No	Three-expert consensus	Open download	NR	Limited harmonized protocol metadata
MS3SEG [30]	100 / 100	Routine MRI; MS lesion versus other WMH	1 site; 1.5 T	No	Junior and senior tri-mask review	Open download	NR	Lesions below 3 mm excluded
PediMS [31]	9 / 28	T1, T2, FLAIR; pediatric lesions	Predominantly one 3-T site	1–6 visits	Specialist-validated masks	Open download	NR	Too small for independent model development
Multicenter cord study [34]	1,849 / 4,428 annotated MRI images	Six cord contrasts; lesions	23 centers; 1.5/3/7 T	Mixed	Expert annotations; external labeled sets	Proprietary/not released (MRI); model public	MRI reuse NR; code separately licensed	Images are the source unit, not independent exams or volumes
MS PATHS [38]	16,568 enrolled; 8,364 with MRI / 14,414 studies	MPRAGE and SPACE-FLAIR; outcomes	10 institutions; Siemens 3 T	Subset	No dense masks at cohort scale	Application-controlled	NR; access agreement applies	Standardized but vendor-specific acquisition
MindGlide sources [40]	Training patients NR (source discrepancy); 4,247 development scans; 1,001 people / 14,952 external scans	Mixed trial and clinical contrasts	592 development / 186 external scanners	Extensive	Trial/cohort-derived labels	Proprietary	NR	Scale limits independent reproduction; patient discrepancy

III. Deep Learning-Based MRI Image Analysis

A. Learning Tasks and Training Paradigms

MS MRI models perform reconstruction, quality control, registration, segmentation, lesion detection, diagnosis, quantitative regression, synthesis, and outcome prediction. Supervised learning dominates because voxel labels define a direct objective, but masks are expensive and inconsistent. Weak supervision uses image- or lesion-level labels; semi-supervision combines labeled and unlabeled data through pseudo-labels or consistency; self-supervision learns representations from transformations or masked content; transfer learning adapts models pretrained on other anatomy or modalities; and federated learning trains across institutions without centralizing images. These paradigms change data requirements, not the need for independent evaluation. Internal cross-validation estimates development stability, whereas site-held-out testing, geographically independent validation, and prospective use answer progressively stronger questions.

The label and sampling unit should follow the task. Reconstruction learns from paired undersampled and reference data or from measurement consistency; quality control may use artifact categories or no-reference objectives; segmentation uses voxel masks but should sample complete patients rather than treating correlated patches as independent; detection can use lesion centers and size; classification needs patient-level diagnoses at a defined decision point; and prognosis needs outcomes anchored after image acquisition. Weak labels are attractive when radiology reports are available, but report absence does not prove lesion absence and reporting style varies by center. Pseudo-labels enlarge training sets while bounding performance to the teacher's systematic errors unless manually adjudicated data remain influential. Consistency learning assumes that perturbations preserve the target, which fails if an augmentation removes a small lesion or if two longitudinal scans contain real change.

Training strategy also determines what can be claimed about missing data. Contrast-specific encoders and modality dropout can support flexible sequence combinations, whereas an early-fusion model assumes all channels are present and accurately aligned. Transfer learning from natural images may initialize edges and optimization; pretraining from other organs or CT provides three-dimensional structure; MS-specific self-supervision can capture scanner and anatomy patterns. None ensures that rare PRLs, cortical lesions, or pediatric appearances are represented. Federated optimization exposes a model to distributed data but does not itself provide a common reference standard. These distinctions should be reported alongside the architecture because two visually identical networks trained under different sampling and supervision regimes may have different failure boundaries.

B. Convolutional and Multiscale Networks

Early deep lesion systems classified patches around candidate voxels. DeepMedic uses parallel three-dimensional paths at native and downsampled resolution, combining local detail with broader context before an optional conditional random field [41]. Patch sampling can balance lesion and background voxels and fit limited memory, but it reduces global anatomical context and produces correlated examples. Fully convolutional networks amortize computation over the image. Two-dimensional, 2.5-D, and three-dimensional implementations trade through-plane context against memory and anisotropy. Dilated convolution expands receptive field without pooling; cascades use a high-sensitivity proposal stage followed by false-positive reduction; and multiscale or multi-arm inputs preserve fine lesions while incorporating global tissue context [42]–[44]. No design eliminates the base-rate problem: many normal voxels and vascular white-matter hyperintensities compete with relatively few, often tiny MS lesions.

C. U-Net and Encoder–Decoder Families

1) U-Net: U-Net accepts a two-dimensional image or multi-sequence slice and couples a contracting convolutional encoder to an expanding decoder [45]. Pooling increases context, upsampling restores resolution, and long skips concatenate—not add—same-scale features. This bias suits dense lesion maps because high-resolution skips retain boundaries while the bottleneck supplies context. Memory depends mainly on image area, feature widths, and scales. Skips can also transmit scanner texture or vascular edges, and imbalance promotes small-lesion omission. Explicit U-Net derivatives and broader fully convolutional systems have MS task-level evaluations [42], [44], [46], but these do not isolate architecture from data, preprocessing, or loss design; the family effect and architecture-specific prospective validation are NR. Fig. 4(a) shows the topology without asserting unreported channels.

2) 3D U-Net: 3D U-Net accepts an aligned volume and replaces two-dimensional operations with volumetric counterparts; its original formulation also assigned zero loss to unlabeled voxels [47]. Three-dimensional receptive fields model interslice continuity for brain, cord, and change maps. Activation storage grows with patch voxels and feature widths, often forcing small patches, coarse resampling, or reduced batches; isotropic kernels may also mismatch thick slices. MS lesion systems exploit volumetric context [43], but the reviewed external comparisons do not isolate canonical 3D U-Net; architecture-specific validation is NR. Fig. 4(b) depicts volumetric operators and voxel output.

3) Residual U-Net: Residual U-Net accepts two- or three-dimensional images and replaces convolution blocks with units that learn $y = F(x) + x$, projecting only when dimensions differ [48]. Short identity addition improves gradient flow, whereas long U-Net skips still concatenate same-scale features. V-Net is a volumetric residual encoder–decoder; R2U-Net adds recurrent convolution within residual units [49], [50]. Greater depth, width, and projections raise memory and overfitting risk, and block recurrence is not longitudinal modeling. No reviewed external MS comparison isolated the residual connection under fixed data and training, so architecture-specific validation is NR. Fig. 4(c) distinguishes addition from concatenation.

4) Dense U-Net: Dense U-Net accepts two- or three-dimensional MRI and places dense blocks within an encoder–decoder. Inside each block, every layer receives the channel-wise concatenation of all preceding feature maps, enabling feature reuse and short gradient paths [51]. This mechanism can preserve multiscale lesion cues under limited training data, but channel growth is memory-intensive and can retain site-specific low-level texture. H-DenseUNet combines two-dimensional dense features with a three-dimensional branch, although its primary evidence arose from liver CT rather than MS MRI [52]. It therefore explains a useful inductive bias but not MS performance. For MS lesion or tissue segmentation, architecture-specific external validation is NR in the reviewed evidence. Fig. 5(a) exposes the all-to-later concatenations; none is drawn as addition.

5) Attention U-Net: Attention U-Net accepts two- or three-dimensional MRI and inserts a gate on each long encoder skip. A coarse decoder signal and the encoder feature jointly produce spatial coefficients; element-wise multiplication then forms the gated skip before decoder concatenation [53]. This mechanism can suppress irrelevant tissue and vascular candidates while retaining fine features for lesion segmentation. It is not Transformer self-attention because it does not construct token-to-token interactions. Resource use depends on the number and spatial size of gated skip maps, and a gate may suppress low-contrast or very small lesions under domain shift. The reviewed sources did not verify an external MS study isolating Attention U-Net from other design choices; MS-specific architecture validation is therefore NR. Fig. 5(b) identifies the gating signal, coefficient, multiplication, and gated skip explicitly.

6) nnU-Net: nnU-Net accepts a labeled segmentation dataset rather than a prescribed image shape and is not a fixed new layer or backbone. It derives a dataset fingerprint and uses rule-based planning plus cross-validation to configure spacing, normalization, patch and batch size, 2-D or 3-D U-Nets, cascades, augmentation, ensembling, and postprocessing [54]. Its advantage is reproducible adaptation of a strong local multiscale pipeline with little manual architecture tuning; resource use depends on planned patch and batch sizes, the number of folds and candidate models, and whether an ensemble is retained. It remains vulnerable to uncertain labels, time-point leakage, and an unrepresentative cohort. MS use of configured pipelines does not by itself isolate a framework effect, and prospective MS-specific validation of nnU-Net as a complete procedure is NR. Fig. 5(c) therefore places candidate U-Nets inside the complete planning and validation process

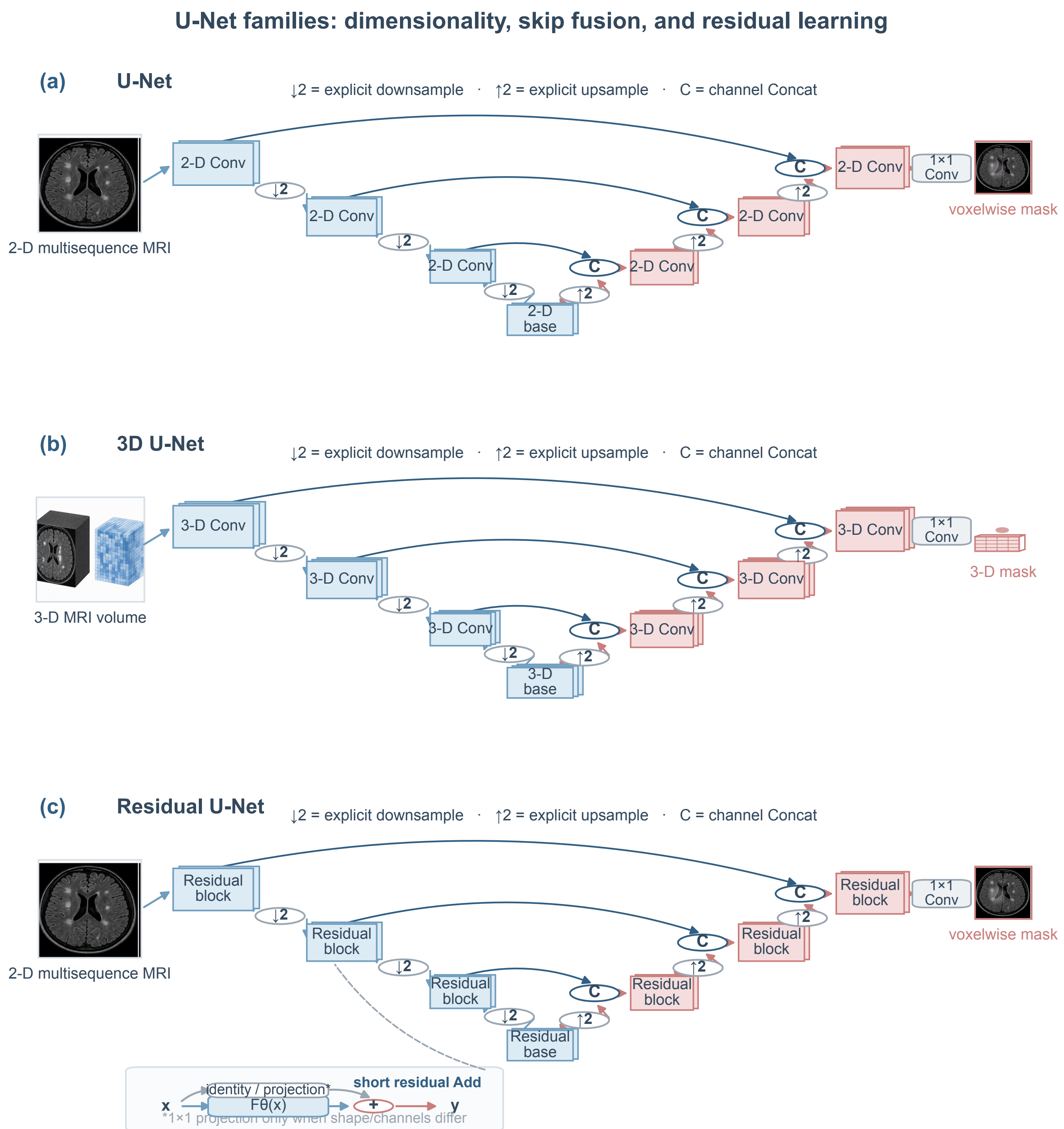

Fig. 4. U-Net, 3D U-Net, and residual U-Net. Pale blue denotes encoder features, pale red decoder features and predictions, and gray neutral operators. Long matching-resolution U-Net skips use channel concatenation (C/Concat). The residual block instead uses a short identity or projection branch with element-wise addition (+/Add). Arrows run from input to prediction.

rather than portraying nnU-Net as a convolution block.

Across these families, architectural labels alone do not predict clinical performance. A shallow 2-D model may be preferable for thick-slice routine imaging; a 3-D residual model may exploit isotropic research scans; attention may reduce false positives but also lower sensitivity; and a self-configured pipeline may fit its development data while remaining externally untested. Table III compares the required input, local or global mechanism, concrete resource determinants, likely advantage, failure mode, and verified MS-specific validation record. Same-dataset comparisons are informative; cross-dataset ranking by Dice is not.

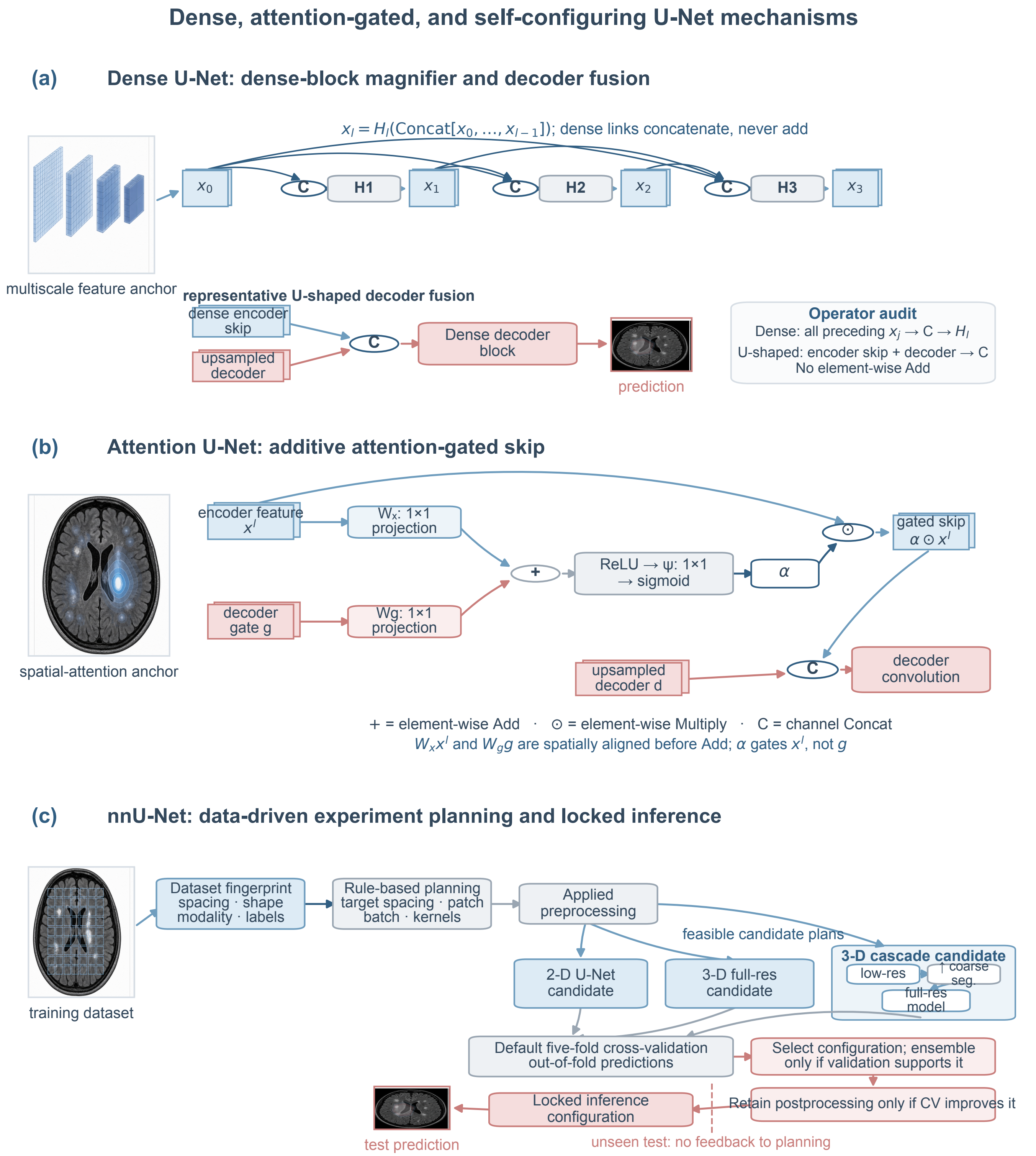

Fig. 5. Dense U-Net, attention U-Net, and nnU-Net. A dense block concatenates all preceding features into each later layer; it does not add them. The attention gate combines an encoder feature with a decoder gating signal to estimate an attention coefficient, multiplies that coefficient with the skip feature (⊙/Multiply), and then concatenates the gated skip in the decoder. nnU-Net is shown as rule-based configuration from a dataset fingerprint through preprocessing, 2-D/3-D/cascade planning, cross-validation, ensembling, and validation-selected postprocessing, not as a novel convolutional block.

D. Deep Unfolding and Data-Consistency Networks

Deep unfolding maps iterations of a model-based optimization algorithm to learnable stages [55]. In MRI reconstruction, a learned regularizer alternates with a data-consistency operation tied to measured k-space and the acquisition operator. MoDL shares a CNN regularizer across stages, while end-to-end

TABLE III
DEEP LEARNING ARCHITECTURE FAMILIES

Family	Input	Core mechanism	MS MRI task	Resource determinants	Principal advantage	Typical failure	MS-specific validation record used here
U-Net	2-D	Encoder–decoder; matching-scale Concat skips	Brain lesion segmentation	Pixel area, feature widths, and number of scales	Preserves local detail	No through-plane context; texture shortcuts	MS task use reported; isolated family effect: NR
3D U-Net	3-D volume or patch	Volumetric operators and Concat skips	Brain and cord segmentation	Patch voxels, feature widths, and activation storage	Models volumetric continuity	Patch context and anisotropy	Canonical 3D U-Net MS external validation: NR
Residual U-Net	2-D/3-D	Short identity Add paths inside blocks	Deeper segmentation	Block depth, feature widths, and projection paths	Improves gradient propagation	Parameter cost and overfitting	Isolated MS validation: NR
Dense U-Net	2-D/3-D	All-preceding feature Concat	Lesion segmentation	Growth rate and accumulated feature channels	Feature reuse	Channel growth and texture dependence	Isolated MS validation: NR
Attention U-Net	2-D/3-D	Decoder-gated encoder skips with $\odot$/Multiply	Small-lesion localization	Number and spatial size of gated skip features	Suppresses irrelevant skip regions	Can suppress subtle true lesions	Attention U-Net specifically in MS: NR
nnU-Net	2-D/3-D/cascade	Dataset fingerprint and rule-based pipeline planning	General segmentation baseline	Planned patch/batch size, folds, candidate models, and ensemble	Reproducible configuration baseline	Cannot repair biased data or labels	Framework-specific MS external validation: NR
Vision Transformer	2-D/3-D tokens	Global MHSA with positional encoding	Classification or encoder backbone	Token count (quadratic attention), width, depth, and heads	Long-range interaction	Data demand and coarse-token detail loss	Stand-alone ViT in MS: NR
Swin Transformer	2-D/3-D tokens	W-MSA, SW-MSA, and patch merging	Dense prediction	Window size, stage depth, token count, and feature width	Hierarchical scalable context	Window and merging losses	Swin-specific MS external validation: NR
UNETR	3-D volume	Transformer encoder; multidepth CNN decoder skips	Volumetric segmentation	3-D token count, encoder depth/width, and decoder activations	Global 3-D context	Memory and token-resolution limits	UNETR-specific MS external validation: NR
CNN–Transformer hybrid	2-D/3-D	Local CNN features followed by global token modeling	Segmentation and classification	CNN feature maps plus token count, width, and depth	Combines local and global bias	Downsampling and token bottleneck	TransUNet/CoTr-specific MS external validation: NR

variational networks learn regularization and coil-sensitivity components around repeated consistency updates [56], [57]. This structure constrains reconstruction more than an unconstrained image generator, but does not guarantee preservation of MS lesions, CVS, or PRLs. Downstream lesion sensitivity, volume bias, and reader assessment should accompany PSNR, SSIM, and NMSE. Fig. 8(a) shows the recurring learned-prior/data-consistency cycle.

E. Transformer and Hybrid Architectures

1) Vision Transformer: Vision Transformer (ViT) accepts a two-dimensional image or three-dimensional volume partitioned into nonoverlapping patches. Linear embedding and positional encoding convert patches into tokens, which pass through pre-normalized multihead self-attention and multilayer perceptron residual blocks [58]. Global token interaction can relate distant lesion locations and diffuse tissue patterns, making ViT a candidate backbone for classification, prognosis, or a dense decoder. It is not itself a complete medical segmentation network. Coarse patches can dilute a small lesion within a mostly normal token, whereas fine volumetric patches create quadratic attention and high memory cost. In the MS-specific evidence reviewed here, robust independent external validation of a stand-alone ViT is NR. Fig. 6(a) therefore ends at encoded tokens and a task-specific head.

2) Swin Transformer: Swin Transformer accepts patch tokens and restricts attention to local windows, alternating window multihead self-attention with shifted-window attention so information crosses previous boundaries [59]. Patch merging reduces resolution while expanding channels, producing hierarchical features suited to lesion segmentation, tissue quantification, and classification heads. Compared with global ViT, this structure lowers attention cost and recovers multiscale organization; window boundaries and merging can nevertheless weaken isolated small-lesion signals, and volumetric variants remain computationally demanding. Self-supervised Swin evidence from large CT collections explains transfer potential but is not MS validation [60]. Independent MS-specific external maturity is therefore NR in the included evidence. Fig. 6(b) shows both window regimes and patch merging rather than generic attention.

3) UNETR: UNETR accepts a three-dimensional volume, forms 3-D patch tokens, and processes them with a single-resolution Transformer encoder [61]. Hidden states from several encoder depths are reshaped and projected to multiple spatial scales, then joined to a CNN decoder. The resulting global-to-local mechanism is appropriate for volumetric lesion and tissue segmentation, but coarse tokenization may erase punctate lesions and fine tokenization imposes substantial GPU memory. Pretraining and augmentation can reduce data demands without resolving domain shift. The reviewed evidence did not verify MS-specific external validation of UNETR; that field is therefore NR rather than a maturity claim. Fig. 7(a) identifies the 3-D tokens, hidden-state taps, and convolutional decoder without adding a CNN encoder or Swin-style patch merging.

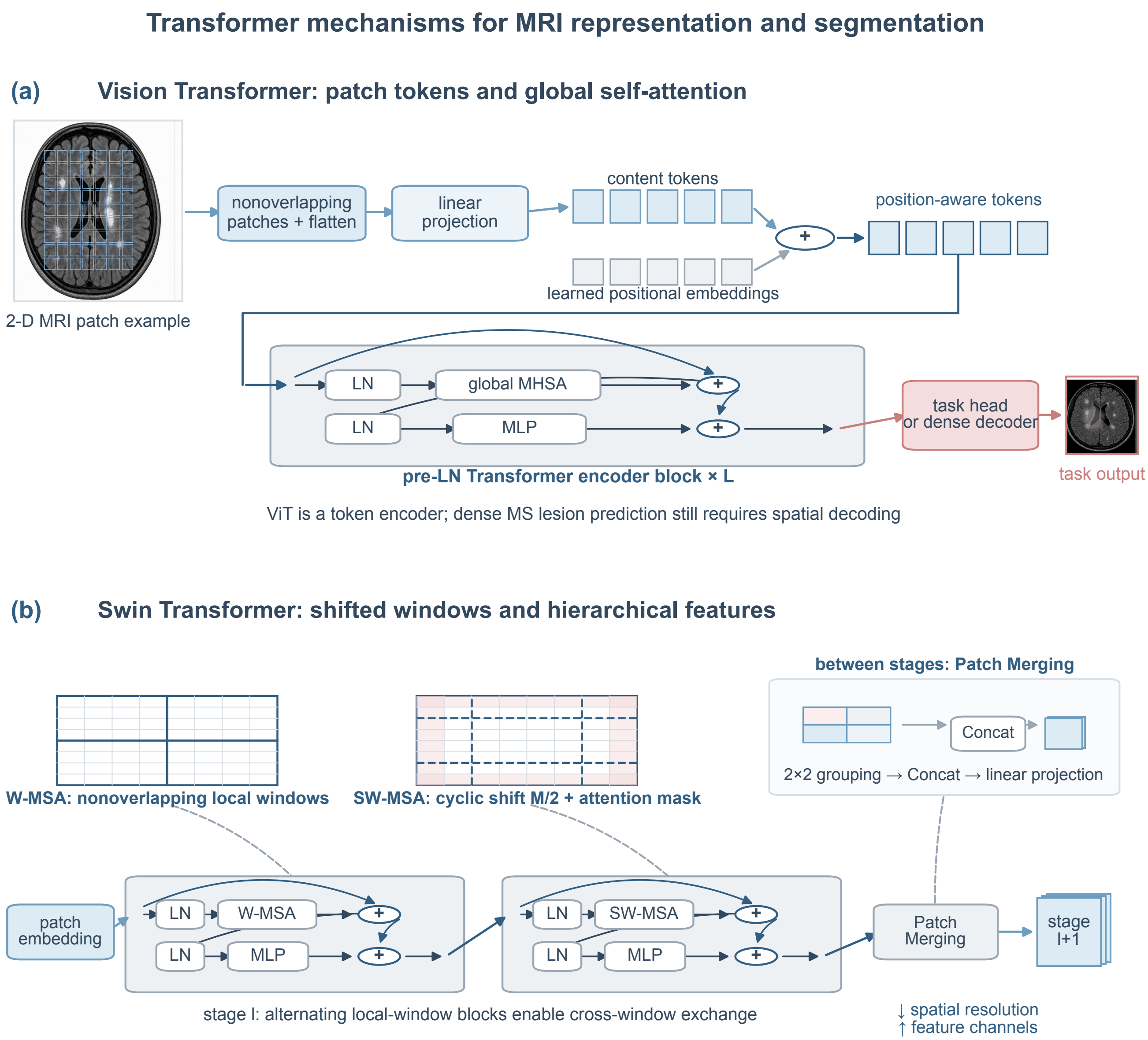

Fig. 6. Vision Transformer and Swin Transformer. ViT partitions an image or volume into patches, applies linear embedding and positional encoding, and repeats pre-normalized MHSA and MLP sublayers with residual addition before a task-specific head or decoder. Swin alternates window attention (W-MSA) and shifted-window attention (SW-MSA) and uses patch merging to form hierarchical features. Gray nodes are token or normalization operations; pale blue blocks are encoders.

4) CNN–Transformer hybrids: A hybrid accepts slices or volumes and orders local and global modeling explicitly. TransUNet first extracts local CNN features, tokenizes the low-resolution map, applies global Transformer encoding, and reconstructs a segmentation with an upsampling decoder and high-resolution CNN skips [62]. CoTr instead applies deformable attention to multiscale three-dimensional CNN features at sparse learned locations [63]. CNN locality helps preserve texture and edges, whereas token interaction expands anatomical context for segmentation or classification. Resource use combines CNN activation maps with the token count, Transformer width, and depth, and small lesions may disappear during CNN downsampling or sparse token selection. The primary TransUNet and CoTr demonstrations were not MS studies; MS-specific external validation of these named hybrids is NR. Fig. 7(b) depicts the verified TransUNet order and does not relabel every attention-equipped CNN as a Transformer.

Comparative claims require restraint. ViT offers global attention but weak inherent hierarchy; Swin provides local windows and hierarchical scaling; UNETR uses a Transformer encoder with CNN decoding for 3-D segmentation; and TransUNet moves from CNN features to tokens before decoding. Self-supervised Swin pretraining has improved medical tasks, but prominent demonstrations used thousands of CT volumes, not MS MRI, so transfer must be revalidated [60]. Their value in MS depends more on lesion scale, data volume, acquisition

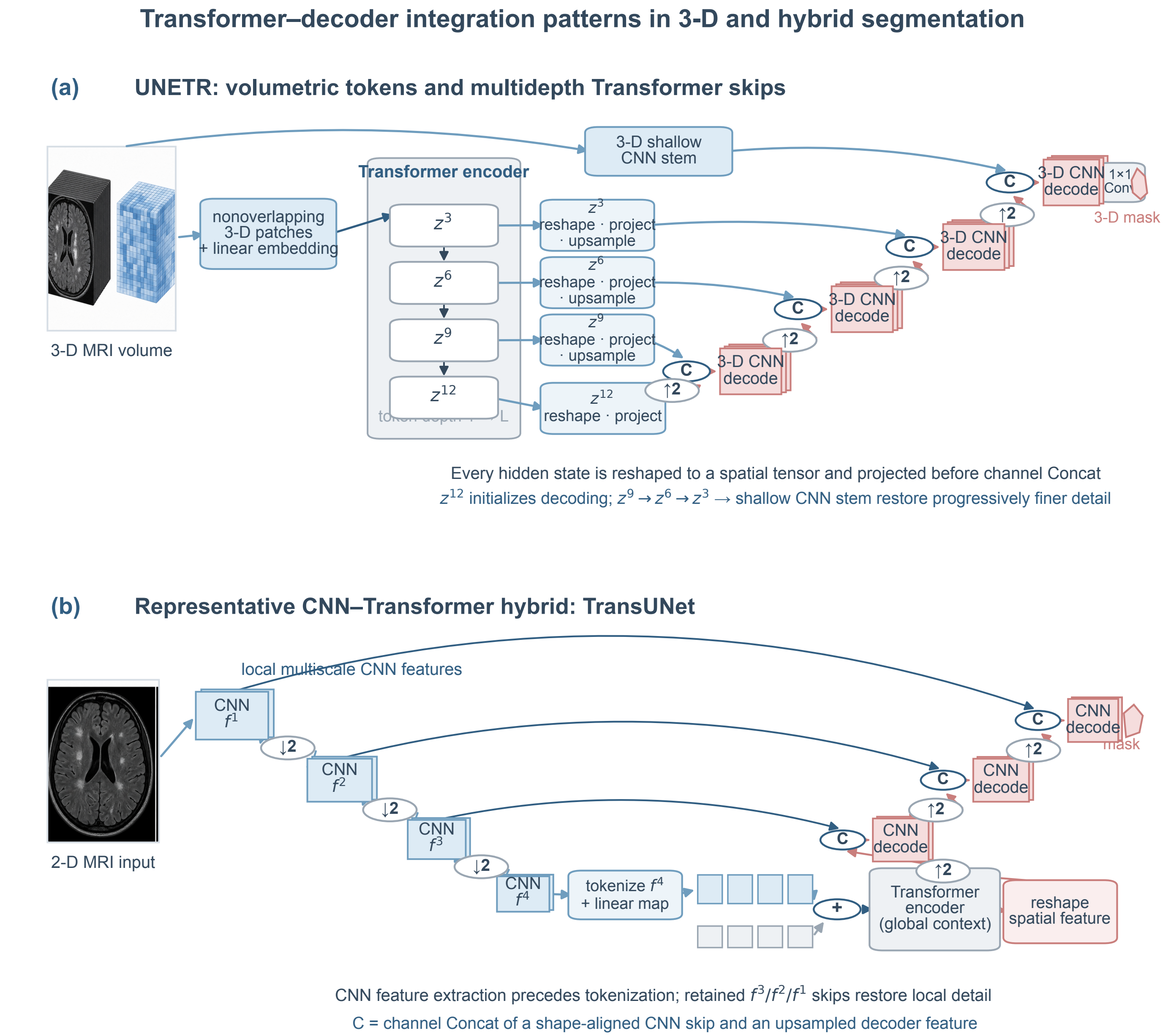

Fig. 7. UNETR and a CNN-Transformer hybrid represented by TransUNet. UNETR embeds 3-D patches, extracts hidden states at several Transformer depths, reshapes them into spatial features, and joins them to a CNN decoder. TransUNet first extracts local CNN features, tokenizes the deepest feature map for global Transformer modeling, and then decodes with retained higher-resolution CNN skips. Pale blue is encoding, pale red is decoding/output, and C denotes channel concatenation.

diversity, and decoder design than on the Transformer label.

F. Generative and Emerging Models

Generative models address missing contrasts, harmonization, super-resolution, augmentation, and anomaly modeling. A variational autoencoder represents a distribution in latent space and balances reconstruction with a Kullback–Leibler penalty [64]. GANs train a generator against a discriminator; conditional GANs use paired inputs, whereas CycleGAN permits unpaired translation through cycle consistency [65]–[67]. Diffusion models learn a reverse denoising process after a fixed forward noising process, and score-based models generalize this view through stochastic differential equations [68], [69]. Visual realism is not pathology fidelity. Registration error can be learned as disease change, and a normality prior can erase rare lesions. Clinical evaluation must measure lesion addition, removal, and modification, not only distributional similarity.

State-space models provide a different long-context mechanism. S4 efficiently represents structured state dynamics; Mamba makes state updates input-selective without query–key–value attention [70], [71]. SegMamba and Swin-UMamba combine convolutional or U-shaped spatial processing with selective state-space blocks [72], [73]. Linear sequence scaling is attractive for three-dimensional volumes and long follow-up, yet spatial flattening order, directional bias, and limited MS-specific external evidence prevent claims of maturity. Fig. 8 separates this selective-state mechanism from probabilistic generation and reconstruction unfolding.

G. Multimodal and Longitudinal Models

Multimodal fusion may occur at the input, within contrast-specific feature encoders, through cross-attention, or after independent decisions. Early fusion is simple but assumes aligned, complete contrasts; late fusion tolerates missing channels but may discard cross-contrast detail. Longitudinal models face a related choice. Two visits can be stacked, processed by separate paths, encoded with shared weights, or represented as a time sequence. Attention-guided two-path networks suppress stable lesions, large three-dimensional CNNs learn new-or-enlarging lesion maps directly, and a verified Siamese design registers baseline and follow-up inputs before shared-weight encoding, feature comparison, and change-map decoding [74]–[76]. The shared weights enforce comparable representations but do not correct registration error; independent segmentations followed by subtraction additionally amplify threshold variation. Self-supervised change detection can create pseudo-change examples, but synthetic changes do not substitute for real scanner drift and treatment effects [77]. For EDSS, PIRA, cognition, or treatment response, time zero, outcome horizon, censoring, relapse, and therapy changes must be specified [78], [79]. Fig. 8(d) makes the temporal direction and shared weights explicit, completing the comparison of emerging architectures.

H. Self-Supervised and Foundation Models

Self-supervision can reduce dependence on lesion masks. Models Genesis reconstructs volumes after intensity and spatial corruption; SimCLR and MoCo learn invariance between augmented views; masked autoencoders encode only visible patches; and DINO trains a momentum teacher without explicit negative pairs [80]–[84]. In MS MRI, an “invariant” augmentation must not remove or relocate a lesion, and same-patient visits are not automatically positive pairs because true disease change may exist. Pretraining from CT or natural images may transfer anatomy and optimization but not sequence-specific pathology.

Promptable foundation models introduce a different interface. Segment Anything combines a reusable image encoder, prompt encoder, and mask decoder; MedSAM adapts this design on a large collection of medical image–mask pairs, commonly using box prompts [85], [86]. Neither is an autonomous MS lesion detector, and two-dimensional prompt performance does not establish volumetric consistency, small-lesion coverage, or longitudinal reliability. Claims of “foundation” capability require transparent training composition, MS-specific external testing, and realistic prompts generated without access to the reference mask.

I. Loss Functions

The training objective determines which errors are cheap or expensive. For reconstruction, the mean absolute error $L_1 = \frac{1}{N}\sum_i |x_i - \hat{x}_i|$ is less dominated by outliers than $L_2 = \frac{1}{N}\sum_i (x_i - \hat{x}_i)^2$; Huber loss interpolates between them. SSIM-based, perceptual, and adversarial terms improve structural or visual appearance, while data-consistency loss penalizes disagreement with acquired measurements. None directly guarantees preservation of an unlabeled lesion, so a downstream task or pathology constraint is needed.

For binary segmentation, cross-entropy is

$$L_{\mathrm{CE}} = -\sum_i \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right], \qquad (1)$$

where $y_i \in \{0, 1\}$ and p_i is the predicted lesion probability. Because background dominates MS volumes, cross-entropy alone may underweight lesions. Soft Dice instead optimizes overlap,

$$L_{\mathrm{Dice}} = 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}, \qquad (2)$$

with a smoothing constant ϵ; implementations must define empty-mask behavior. Generalized Dice reweights classes by inverse volume [87]. A smoothed Tversky loss is

$$L_{\mathrm{T}} = 1 - \frac{\mathrm{TP} + \epsilon}{\mathrm{TP} + \alpha\mathrm{FP} + \beta\mathrm{FN} + \epsilon}. \qquad (3)$$

where TP, FP, and FN are soft true-positive, false-positive, and false-negative sums; α and β are nonnegative error weights, and ϵ is the smoothing constant. Setting $\beta > \alpha$ penalizes missed lesion voxels more strongly. This loss was introduced for highly imbalanced three-dimensional MS lesion segmentation [88]. Focal loss multiplies cross-entropy by $(1 - p_t)^\gamma$, where p_t is the probability assigned to the true class and $\gamma \geq 0$ is the focusing parameter; it emphasizes difficult examples while also amplifying mislabeled voxels [89]. Dice or Tversky loss can improve sensitivity without ensuring realistic lesion counts.

Boundary loss integrates predicted probabilities with the signed distance map of the reference contour; Hausdorff-oriented objectives provide differentiable approximations to extreme boundary error [90], [91]. They are usually combined with regional loss because a boundary term cannot recover a lesion missed completely. “Surface loss” should not be treated as one universal formula, and the surface Dice metric is not automatically a training loss. For representation learning, contrastive losses separate or align embeddings, masked reconstruction predicts hidden content, and mean-teacher consistency matches student and exponential-moving-average teacher outputs under perturbation [92]. Prognosis needs outcome-aware objectives: regression loss fits continuous scores, ranking loss orders risk, and Cox partial likelihood handles censoring through risk sets [93], [94]. DeepHit offers a non-proportional-hazards formulation for discretized competing risks [95]. Fig. 9 links four clinically recognizable segmentation errors—small-lesion omission, false-positive candidates, boundary displacement, and volume bias—to complementary regional, asymmetric, focal, and boundary-aware objectives. It does not imply that any loss

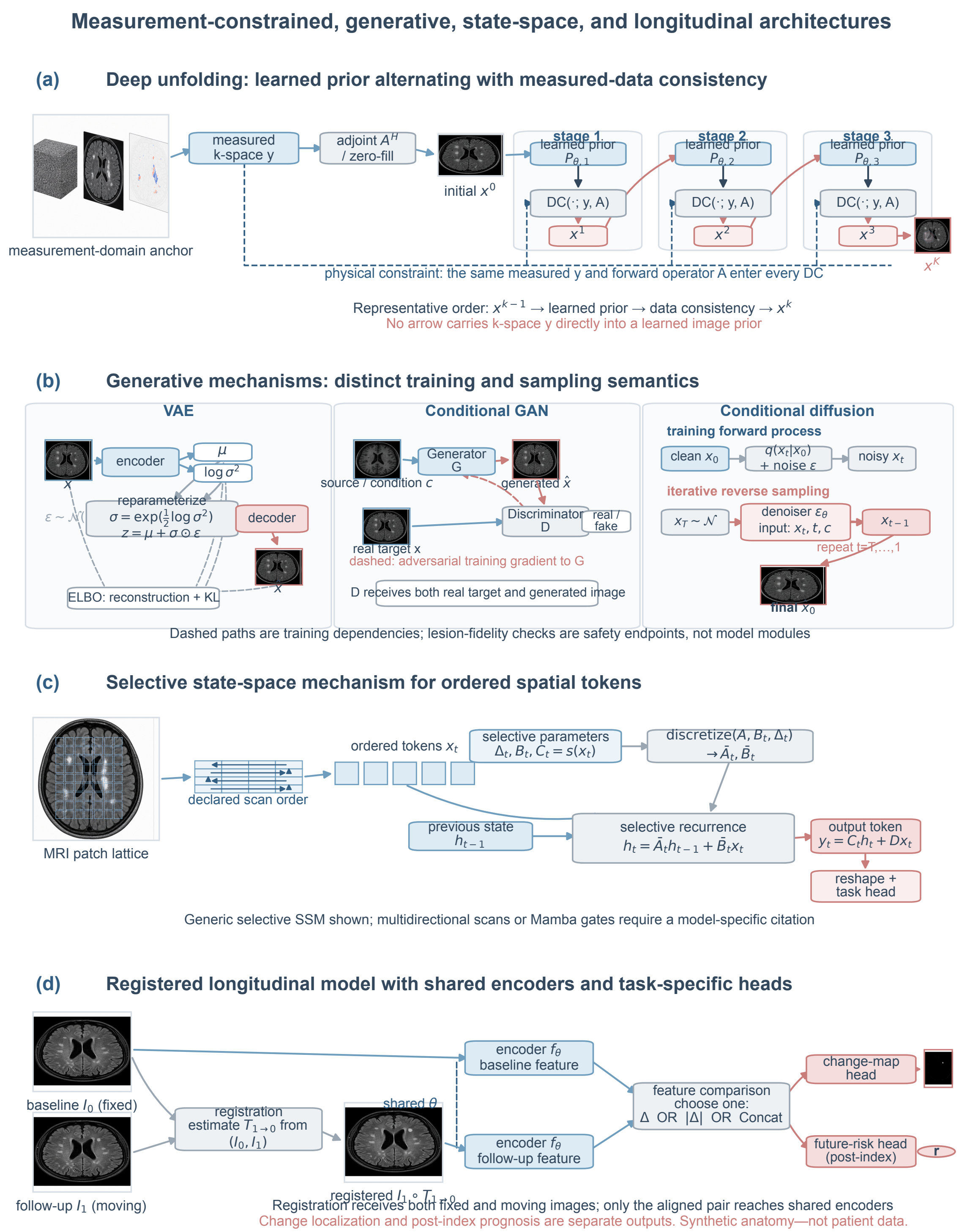

Fig. 8. Emerging and longitudinal architectures. Each deep-unfolding stage contains a learned prior followed by a data-consistency operation constrained by the same measured k-space. Separate panels distinguish VAE, GAN, and diffusion mechanisms; selective state-space scans; and longitudinal analysis with baseline and follow-up registration, shared-weight encoders, feature comparison, and change-map or risk output. The MRI textures are synthetic schematic anatomy, not patient data.

guarantees correction. Table IV extends this task–error mapping to reconstruction, representation learning, and prognosis.

J. Metrics and Evidence Quality

Reconstruction metrics include PSNR, SSIM, and NMSE, but lesion preservation and downstream measurement error are more clinically meaningful. Segmentation requires Dice

Fig. 9. Clinically distinct segmentation errors require different objective emphasis and reporting bundles. Synthetic FLAIR thumbnails with vector contours illustrate small-lesion omission, a false-positive lesion, boundary displacement, and volume bias. Focal, Tversky, specificity-aware overlap, regional, boundary, surface, and volume-aware terms address different error components, but no loss guarantees lesion recovery. No empirical performance values are shown.

together with lesion-wise sensitivity, positive predictive value, false-positive lesions per scan, volume bias, and HD95. A high Dice can coexist with missed small lesions because large confluent lesions dominate voxels. Classification requires sensitivity, specificity, predictive values at realistic prevalence, AUC, calibration intercept and slope, and decision-curve analysis. Prognosis requires C-index or time-dependent AUC plus calibration, Brier score, and net benefit [96], [97]. Metric bundles should therefore be selected from the target error and intended decision rather than from convention. Fig. 10 links five complementary requirements: patient-level separation and locked external testing; overlap, lesion, boundary, and burden operators for segmentation; case-spectrum and time-context constraints on diagnostic and prognostic claims; stratified aggregation with uncertainty and abstention; and an evidence ladder that bounds the permitted clinical role. Metric adequacy does not upgrade a study from internal to external or prospective validation, nor does it establish clinical utility. The figure contains no curves or empirical performance values.

Evidence strength rises from subject-level internal validation to a same-institution hold-out, leave-site-out testing, geographically independent external validation, temporal validation, prospective workflow study, and patient-outcome evaluation. Domain-shift and uncertainty studies show that confident segmentation can fail on unfamiliar scanners; calibration and out-of-distribution detection must be assessed together [98]–[100]. Large-scale MS systems and open tools demonstrate that multicenter training is feasible, but external performance commonly falls relative to development data [46], [101], [102]. Code, weights, preprocessing, thresholds, and patient-level

TABLE IV
LOSS FUNCTIONS AND TASK-SPECIFIC METRICS

Objective family	Examples	Best-matched task	Strength	Failure mode and required metric
Voxel reconstruction	L1, L2, Huber	Reconstruction and synthesis	Stable intensity fidelity	Blur or outlier sensitivity; add SSIM, NMSE, and lesion-preservation tests
Perceptual / adversarial	Feature and GAN losses	Synthesis and harmonization	Plausible texture	Hallucination; adjudicate lesion addition and deletion
Data consistency	Measured k-space residual	Accelerated reconstruction	Preserves acquired measurements	Does not guarantee pathology; evaluate downstream tasks
Voxel classification	Cross-entropy and focal	Segmentation and detection	Probabilistic local target	Imbalance and noisy labels; add lesion sensitivity and PPV
Regional overlap	Dice, generalized Dice, Tversky	Imbalanced segmentation	Optimizes overlap	Volume dominance and empty masks; add lesion-instance metrics
Boundary geometry	Boundary and Hausdorff-oriented	Contour refinement	Penalizes shape error	Cannot recover a completely missed lesion; report HD95
Representation	Contrastive and masked reconstruction	Self-supervised pretraining	Uses unlabeled images	Invalid invariances; test downstream external data
Consistency	Teacher–student and pseudo-label	Semi-supervised learning	Uses unlabeled images	Confirms teacher errors; audit calibration
Outcome	Regression, ranking, Cox, DeepHit	EDSS, PIRA, and cognition	Handles continuous or censored targets	Confounding; report calibration, C-index, and net benefit

splits are all needed for reproducibility. MI-CLAIM and the medical algorithmic-audit framework complement task-specific reporting by making the model, data lineage, and lifecycle testable [103], [104].

The validation unit must match the clinical claim. Voxel-level confidence intervals do not account for correlation within lesions, patients, and centers. Cross-validation folds should be grouped by patient, and hyperparameter selection should be nested so that the reported test fold does not influence model choice. A "multicenter" random hold-out may still contain images from every training institution and thus measures interpolation within a shared ecosystem. Leave-one-site-out testing is stronger, but coordinated trials can share eligibility, protocol, and central preprocessing. A geographically and operationally independent PACS cohort tests a more meaningful shift; prospective temporal ordering additionally tests software and protocol drift. Reader studies should randomize or counterbalance AI assistance, use current best practice as the comparator, and capture corrections, confidence, time, missed activity, and downstream recommendations.

Interpretability should be evaluated as an interface and safety mechanism, not as a colorful explanation appended to an invalid model. Saliency maps may identify borders, skull stripping, ventricles, or lesions, but are unstable and do not establish causal reasoning. More actionable mechanisms include input-quality checks, explicit sequence recognition, lesion-level candidates linked across time, editable contours, calibrated uncertainty, and abstention when the case is outside the supported domain. The output must distinguish "no lesion detected" from "analysis unreliable." Reporting should specify who reviews the result, how disagreement is resolved, where failures are logged, and whether an update changes thresholds or preprocessing. These details determine the behavior of the human–AI system more directly than a small improvement in mean Dice.

K. Task-Specific Evidence and Clinical Translation

1) White-matter and enhancing lesions: Cross-sectional brain white-matter lesion segmentation is the most extensively benchmarked task. Large-scale fully convolutional studies, simultaneous tissue–lesion segmentation, open longitudinal pipelines, BIANCA-MS, and nnU-Net-style systems commonly show useful overlap and volume agreement under matched acquisition [42], [43], [101], [102]. The clinically difficult tail is consistent across methods: small, juxtacortical, infratentorial, low-burden, and confluent lesions; scans with vascular white-matter hyperintensities; and unfamiliar scanners. Because Dice is volume-weighted, a model can achieve an acceptable patient mean while missing several small lesions. Lesion sensitivity, positive predictive value, false positives per scan, case-level activity classification, and manual correction time are therefore indispensable. Contrast-enhancing lesions are rarer and more imbalanced. Multicenter and clinical-MRI studies report technically useful segmentation, but vessels, motion, timing after contrast, and punctate lesions remain major error sources [105], [106]. A high voxel overlap on selected positive cases is not evidence that a model will behave safely in a population where most scans contain no enhancement.

2) Spinal, cortical, and susceptibility lesions: Cord segmentation can reach high geometric agreement because the cord is a relatively continuous structure; intramedullary lesion segmentation is substantially harder because the target is small, contrast varies by level and sequence, and motion and partial volume are prominent [33], [34]. A deployable system should separate failure of cord localization from failure of lesion detection and should flag incomplete axial coverage. Cortical-lesion models have benefited from 7-T and specialized contrasts, but performance declines across field strength and institution, especially for intracortical lesions [107], [108]. CVS and PRL systems illustrate the gap between lesion-level classification and end-to-end patient assessment. CVSnet achieved promising results after candidate lesions were selected, while RimNet and QSMRim-Net address specialized susceptibility inputs and severe class imbalance [109]–[111].

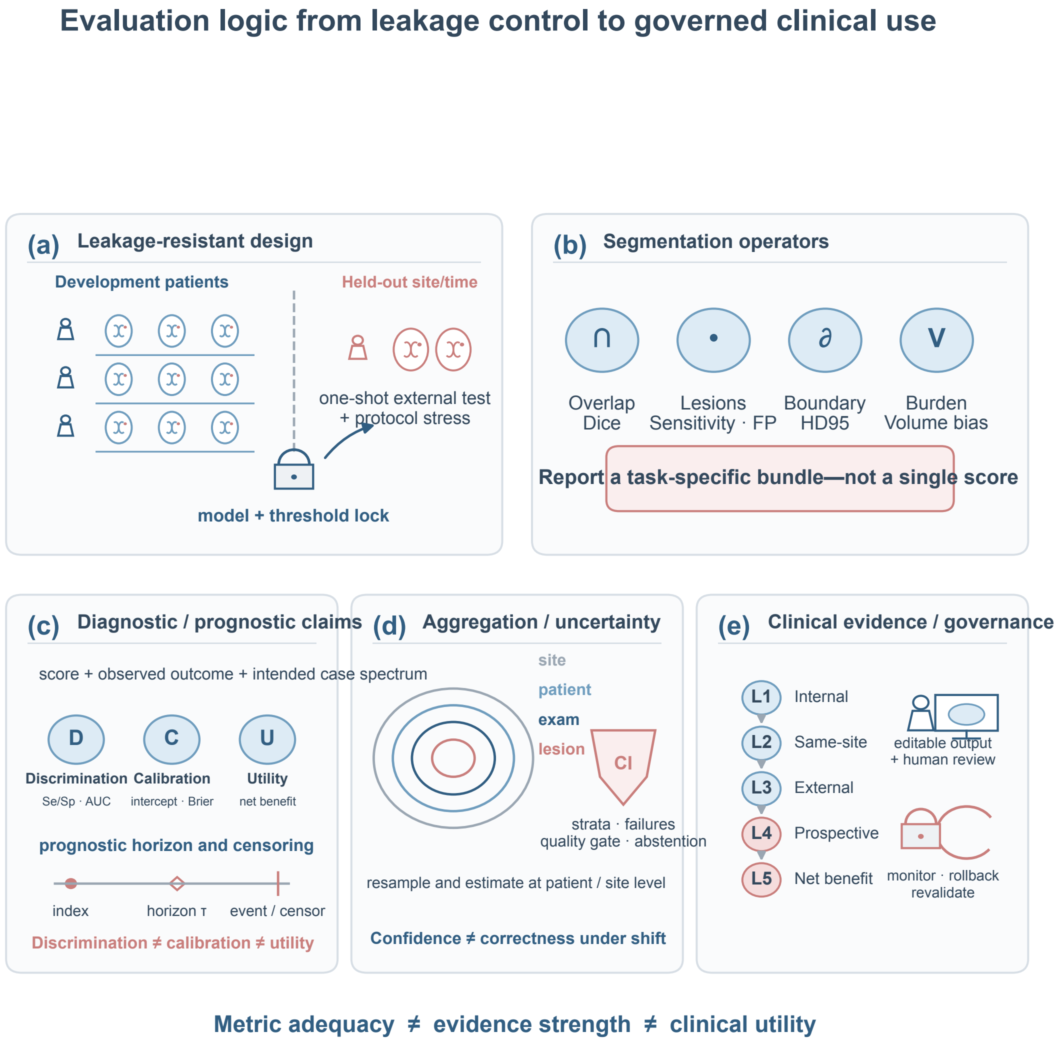

Fig. 10. Evaluation logic from leakage control to governed clinical use. Five panels connect patient-level splitting and locked external testing, complementary segmentation metrics, bounded diagnostic and prognostic claims, stratified aggregation with uncertainty and abstention, and an evidence ladder from internal testing to patient net benefit. Pale blue denotes technical measurement and validation operations, pale red denotes clinical or governance actions, and gray denotes neutral constraints. Metric adequacy does not establish evidence strength or clinical utility. No empirical performance values are shown.

A clinically credible pipeline must find eligible lesions, reject non-evaluable candidates, aggregate patient-level rules, and test the entire process in a suspected-MS spectrum. Combined automatic segmentation of lesions, veins, and rims is an important direction but remains dependent on acquisition quality and external replication [112], [113].

3) Longitudinal monitoring and quantitative measurement: Detecting new or enlarging lesions is closer to the clinical question of inflammatory activity than segmenting total burden. Direct paired-input networks avoid subtracting two independently thresholded masks, but remain sensitive to registration, intensity drift, slice mismatch, and unavailable priors. MSSEG-2 shows that lesion-wise overlap can be modest even when patient activity categories are clinically useful [24]. Imaging-aware augmentation, large private multiscanner training, and PACS-compatible systems improve robustness, and reader studies suggest that candidates can reduce reporting time or increase activity detection [75], [114]–[118]. These gains are conditional: false positives may transfer time from search to verification, and a previous report is not an adequate substitute for image-level comparison. Brain-volume and tissue segmentation offer another quantitative pathway. Longitudinal volume change requires test–retest stability, bias and limits of agreement, scanner-upgrade analysis, and consistent intracranial-volume handling. High cross-sectional correlation is insufficient, and pseudo-label training can improve consistency while inheriting the reference software's bias [40], [119].

4) Diagnosis, phenotyping, and prognosis: Image classifiers can distinguish established MS from healthy controls or selected mimics, and emerging tools combine CVS, cortical lesions, and PRLs [120]–[123]. The main limitation is not representation capacity but case spectrum. Healthy volunteers often come from a different acquisition source, disease prevalence is artificial, and the model lacks clinical examination, cerebrospinal fluid, serology, and prior probability. Sensitivity and specificity should be measured in consecutive patients referred for suspected inflammatory disease, with calibration and decision curves at clinically relevant thresholds. Prognostic studies use lesion burden, tissue volume, radiomics, or deep features to predict EDSS, cognition, PIRA, and progression [124]–[129]. Meta-analytic summaries show heterogeneous discrimination and error, but calibration and independent testing are uncommon [130]. Treatment is a time-varying confounder: risk under observed care is not an individual treatment effect. Unsupervised MRI subtypes and latent disease states can be reproducible and biologically informative, as large external replications demonstrate, yet they should not be converted into treatment recommendations without prospective utility evidence [131], [132].

Across these tasks, the same architecture can occupy different maturity levels. A U-Net used to prepopulate editable FLAIR contours may be close to routine assistance; the same backbone used to predict five-year disability may be exploratory because the target, censoring, treatment, and calibration are harder. Conversely, a sophisticated Transformer does not confer clinical validity if tested on a small same-center split. Table III is therefore an architectural comparison rather than a readiness ranking. Table IV connects outputs to appropriate optimization and measurement, whereas Table V retains a numerical result only when the cohort, case condition, metric, and validation level are traceable in the evidence ledger; incompatible rows are not ranked.

IV. Challenges and Future Research

A. Data Diversity and Reference Labels

Public benchmarks are small and geographically concentrated, trial archives are selected, and routine data contain noisy labels and informative missingness. Spinal, cortical, pediatric, progressive, comorbid, and lower-resource populations remain particularly underrepresented. Future resources should report unique patients separately from scans; preserve patient, visit, and site independence; include stable negative examinations; disclose demographics and therapy; and retain hidden scanner-upgrade and external-center test sets. Because reader disagreement concentrates in small or equivocal lesions, a single fused mask should not erase the measurement process. Multirater labels, adjudication rules, probabilistic targets, lesion-size strata, and uncertainty-aware evaluation are needed. Federated learning can broaden institutional participation, but early MS studies also expose unresolved heterogeneity in labels, site contribution, privacy, and update validation [133], [134].

B. Domain Shift and Missing Inputs

Field strength, vendor, protocol, reconstruction, software upgrades, and preprocessing all alter the input distribution. A model should therefore be stress-tested across prespecified acquisition factors, with input quality gates and explicit abstention. Harmonization or synthesis must preserve lesion count, volume, enhancement, CVS, and PRL; visually plausible output is insufficient. Missing sequences create a related risk: complete-case training selects richer sites, zero filling can reveal site identity, and imputation can manufacture disease signal. Controlled modality dropout, performance for each supported sequence subset, and separate imputation and prediction uncertainty are safer. Longitudinal evaluation must additionally account for variable intervals, treatment timing, relapse, registration quality, and mismatched protocols rather than treating these factors as invisible preprocessing.

C. Reliability, Fairness, and Computational Access

Reliability requires more than an uncertainty heat map. Calibration, out-of-distribution detection, and abstention should be linked to a defined review action, and the output must distinguish a negative result from an unreliable analysis. Subgroup assessment is essential because sex and demographic imbalance can become classifier bias [135]. Privacy-preserving training does not automatically produce equitable performance, and site-level aggregates may hide failure in small subgroups. Three-dimensional networks, fine-grained Transformers, ensembles, and foundation models also impose memory, latency, and infrastructure costs that can restrict access. Reports should give accelerator, memory, preprocessing and inference time, failure rate, and supported hardware. Compression or efficient attention should be rechecked for small-lesion sensitivity and calibration; speed without detectable failure is not clinical efficiency.

D. Workflow, Regulation, and Lifecycle Governance

Brain white-matter segmentation and longitudinal change detection have the strongest workflow evidence, including retrospective reader studies, but false-positive verification and acquisition matching still determine net time saved [114]–[118], [136], [137]. Enhancing, spinal, cortical, CVS, and PRL systems more often depend on rare positives, manual candidates, or small external cohorts [105]–[112]. Outputs should be editable, linked to prior images, and accompanied by provenance and failure flags. Regulatory clearance is specific to a product version, input, and intended use; it does not establish improved relapse or disability outcomes. FDA records rechecked on 29 August 2026 list substantially-equivalent 510(k) decisions for Pixyl.Neuro (K213253), icobrain (K192130), and NeuroQuant (K241098), while the NICE MIB291 page remains a briefing rather than evidence of patient benefit [138]–[141]. PACS routing, cybersecurity, local validation, version control, drift monitoring, and a named failure owner are therefore part of the intervention [142]–[144].

E. Prospective Validation and Clinically Meaningful Endpoints

Future studies should begin with the decision, not the architecture. Lesion-monitoring endpoints include actionable

TABLE V
TASK-SPECIFIC TECHNICAL PERFORMANCE AND CLINICAL EVIDENCE

Task and source	Dataset or cohort condition	Reported evaluation	Validation level	Evidence-supported interpretation	Clinical evidence gap
New-lesion benchmark [24]	100 paired-FLAIR patients: 40 development and 60 hidden test; 35 hidden-test cases had new lesions	Among 35 positive hidden cases, best expert mean lesion F1 0.679 (SD 0.345) and best submitted method 0.698 (SD 0.295); across the 60-case hidden test, best three-category accuracy was 85% for both	L2 hidden benchmark with expert comparison	Case classification can be useful while lesion matching remains difficult	Distribution-specific; no prospective workflow or outcome test
MS lesion vs. other WMH [30]	100 patients; one 1.5-T scanner; lesions <3 mm excluded	Five-fold internal U-Net Dice: 0.7469 binary and 0.6686 multiclass	L1 internal cross-validation	Demonstrates feasibility for the resource-defined distinction	Single scanner and lesion exclusion preclude transportability claims
Multicontrast cord lesions [34]	1,849 people; 4,428 annotated MRI images; 23 centers; six contrasts at 1.5/3/7 T	Neuroradiologist Likert comparison favored the proposed model ($p < 0.01$); a single harmonized segmentation estimate is NR here	L3 retrospective multicenter/external evaluation	Tests contrast, center, level, and resolution heterogeneity	Effect size, prospective correction burden, and patient benefit are NR
Federated brain-lesion segmentation [133]	Public and in-house multi-client experiments; patient counts and common sequence set NR in the audited record	Case-wise/voxel-wise Dice: 65.20%/74.30% (public scenario) and 53.66%/62.31% (in-house scenario)	L1 internal multi-client experiments	Shows optimization behavior under decentralized non-IID clients	Not an unseen-center or prospective clinical validation
Routine longitudinal activity [115]	282 patients; 397 scan pairs; principally three scanners/centers	Case-level sensitivity/specificity: AI 93.3%/97.6%, standard radiology 58.3%/98.8%, core laboratory 85.0%/96.4%; stable-protocol subset sensitivity 91.7% for AI and core laboratory	L3 retrospective real-world validation with human comparators	Supports assistive monitoring under the studied conditions	Consensus adjudication and retrospective design limit autonomous or outcome claims
AI-supported reporting time [117]	Four radiologists; 50 lesion-burden studies and 50 follow-up studies; patient overlap NR	Mean time fell from 286.85 to 196.34 s (burden) and 196.17 to 120.87 s (follow-up)	L4 retrospective crossover reader study	Demonstrates a reader-time effect in the tested workflow	Multicenter generalizability and patient outcomes are NR
Automated serial-reading time [118]	35 consecutive single-site patients; baseline plus three follow-ups	Mean reading time 9.05 min; AI assistance reduced it by 2.83 min	L4 retrospective paired-reader study	Suggests workflow efficiency for serial review	Small single-site study; safety and net benefit remain NR
Year-3 PIRA prediction [128]	719 newly diagnosed patients from three Italian centers; 92 developed year-3 PIRA; structured clinical/radiological variables, not image tensors	Random-forest test AUC 0.75 ± 0.06 under nested randomized cross-validation	L1 internal nested cross-validation	Supports cohort-level prognostic association under observed care	External calibration, treatment-confounding control, and decision net benefit are NR
Archive-scale quantitative analysis [40]	4,247 development scans/592 scanners; 14,952 external scans from 1,001 people/186 scanners	Single task-comparable aggregate result: NR	L3 external multiscanner evaluation	Demonstrates evaluation across heterogeneous archives	Patient-count discrepancy and proprietary composition limit reproduction

new activity, false alerts per stable examination, correction burden, reporting time, and treatment review. Volumetry requires test–retest repeatability, annualized bias, quality-control rejection, and incremental value beyond standard reading. Diagnostic support requires consecutive patients with realistic mimics, calibrated post-test probabilities, and consequences of false decisions. Prognosis requires a prespecified time origin, treatment and censoring handling, external calibration, and decision-curve net benefit. Economic analysis should include integration, maintenance, repeat imaging, false-positive work-up, and downstream treatment rather than minutes saved alone [145], [146].

A staged validation program follows from these endpoints. Development should use site-diverse data, patient-level splitting, transparent exclusions, multirater references, and locked thresholds. The frozen system should then be tested at institutions absent from training, including negative scans, protocol stress tests, and demographic and disease subgroups; any recalibration requires a subsequent independent sample. Prospective human–AI evaluation should log quality-control rejection, abstention, edits, disagreement, decision change, and downstream consequences. Randomized or stepped-wedge designs are appropriate when the intervention changes workflow. AssistMS exemplifies this direction by targeting disease-activity detection, treatment decisions, reporting time, and health economics, but its outcomes must not be assumed before completion [147].

Fig. 11 organizes acquisition variability into three research planes: robustness (site-held-out evaluation, protocol stress, missing modalities, and failure localization); trust (multirater uncertainty, subgroup audit, abstention, and transparent review); and translation (locked external validation, human–AI evaluation, net benefit, and monitored rollback). Table VI records the corresponding gaps. Controlled servers, locked federated containers, executable inference code, and versioned model cards can make claims testable without releasing identifiable MRI. The objective is transportable, monitored clinical benefit rather than another marginal leaderboard gain.

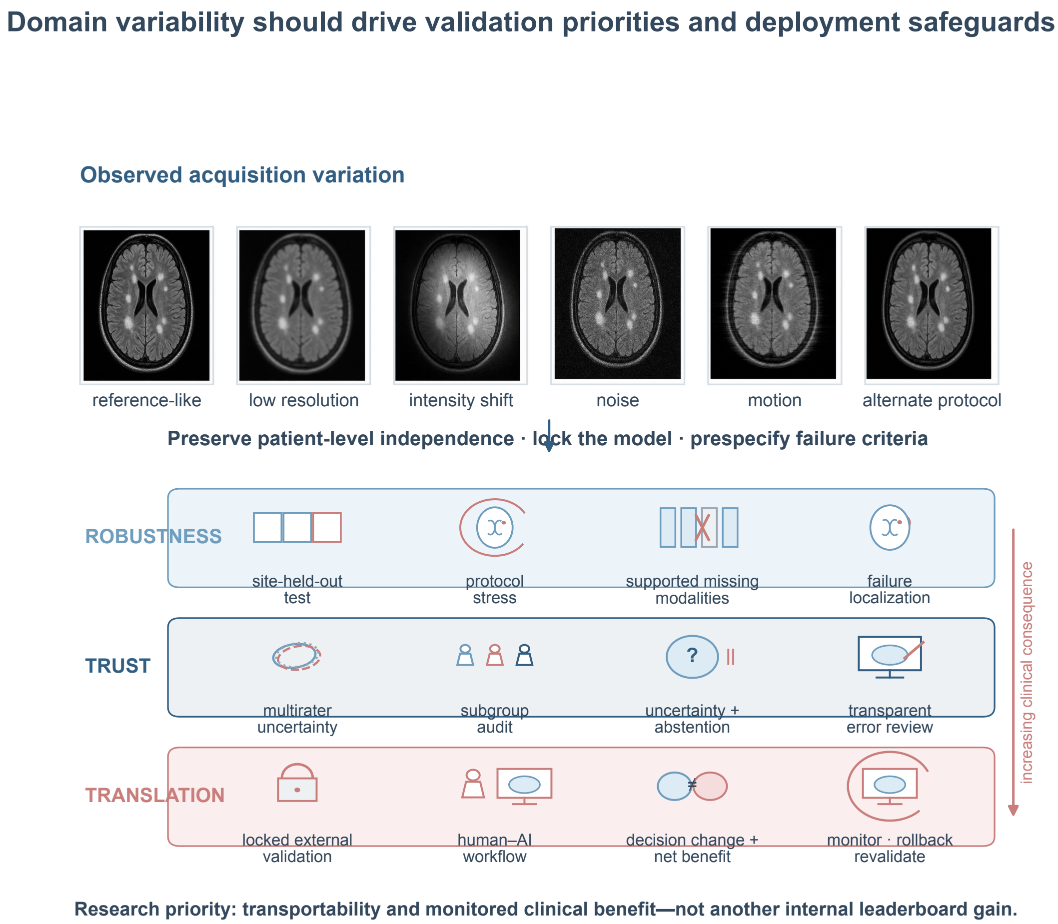

Fig. 11. Domain variability and validation priorities for MS MRI AI. Synthetic FLAIR anatomy illustrates reference-like, low-resolution, intensity-shifted, noisy, motion-degraded, and alternate-protocol inputs; it is not patient data. The three research planes separate robustness tests, trust safeguards, and translation endpoints. The downward arrow denotes increasing clinical consequence, while the individual glyphs specify site-held-out testing, protocol stress, supported missing modalities, failure localization, multirater uncertainty, subgroup audit, abstention, transparent review, locked external validation, human–AI workflow evaluation, net benefit, and governed monitoring or rollback. No empirical performance values are shown.

TABLE VI
CHALLENGES, EVIDENCE GAPS, AND RESEARCH PRIORITIES

Challenge	**Typical evidence gap**	**Priority action**	**Validation requirement**	**Clinical endpoint**
Small or unrepresentative data	Academic, trial, or regional concentration	Consecutive multicenter cohorts	Patient- and site-held-out testing	Subgroup-safe errors
Scanner and protocol shift	Same-ecosystem hold-out	Vendor, field-strength, protocol, and upgrade stress tests	Geographic and temporal external sets	Stable actionable findings
Noisy reference labels	One fused binary mask	Multirater probabilistic reference and adjudication record	Agreement and lesion-size-stratified analysis	Reviewable candidates
Tiny lesions and imbalance	Aggregate Dice only	Lesion-aware sampling and complementary objectives	Negative cases; lesion sensitivity and false positives per scan	Activity detection
Missing or longitudinal inputs	Complete-case or matched protocols	Modality dropout and variable-time models	Every supported sequence subset plus drift analysis	Reliable interval change
Uncertainty and calibration	AUC or Dice without reliability	Calibrated abstention linked to review	External calibration and out-of-distribution tests	Safe manual verification
Privacy and fairness	No subgroup or model-update audit	Federated evaluation and bias monitoring	Site and demographic strata	Equitable performance
Compute and integration	No latency or failure reporting	Efficient models and PACS engineering	End-to-end operational test	Time and correction burden
Clinical utility	Retrospective technical endpoint	Prospective human–AI evaluation	Randomized or stepped-wedge study plus monitoring	Decisions, outcomes, and net benefit

V. Conclusion

AI in MS MRI is closest to clinical assistance for brain-lesion segmentation, quantitative tissue and lesion measurement, and longitudinal detection of new or enlarging lesions, provided that outputs remain editable and are verified by clinicians. Automated enhancing, spinal, cortical, CVS, and PRL analysis has task-specific technical promise but less broadly replicated workflow evidence. Image-only diagnosis, individualized prognosis, generative synthesis, foundation models, and autonomous decision-making remain primarily at technical-validation stage. Their transition should be judged by independent multicenter and prospective human–AI evaluation, not by architecture novelty or internal accuracy alone. The appropriate near-term role of AI is therefore to make measurement and comparison more reproducible while preserving accountable clinical interpretation.

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